

Autumn Examination Semester-2022

Course Code: PHYUGCC01

Course Title: Mathematical Physics-I

Department of Appearing Students: PHYSICS, SEM-I, 1st Year

Full Marks: 60

Time: 2.5 hrs

Group-A

Answer any ten questions

10×2

1. Find the component of $(\hat{i} - \hat{j})$ along $(\hat{i} + \hat{j})$. If a unit vector is represented by $0.5\hat{i} + 0.8\hat{j} + c\hat{k}$, find the value of c.
2. Find the values of (i) $\nabla(\sin r)$ and (ii) $\nabla(\ln r)$.
3. The curl of a vector field $\vec{F}$ is $2\hat{i}$. Find the appropriate vector field $\vec{F}$.
4. Find the eigenvalues of the matrix $A = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$.
5. Two matrices A and B are given below:

$$A = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \text{ and } B = \begin{pmatrix} pq + qs & pr + rs \\ ps + rq & pq + rq \end{pmatrix}$$

If the rank of matrix A, $\rho(A) = n$, what is the value of $\rho(B)$.

6. If $y_1(x)$ and $y_2(x)$ are the solutions of a linear homogeneous second order differential equation, show that the function $y(x) = c_1 y_1(x) + c_2 y_2(x)$ is also a solution of that equation.
7. Show that rank of matrix AB, $\rho(AB) \leq \min [\rho(A), \rho(B)]$.
8. Prove that if A is an invertible matrix, A^T is also an invertible matrix and $(A^T)^{-1} = (A^{-1})^T$.
9. Evaluate $\int_0^3 \delta(t - 2) 2e^{2t^2} dt$.
10. Find the solution of $y'' - 4y' + 4y = 0$.
11. Suppose A and P are square matrices and P is non-singular. Then A and $P^{-1}AP$ have the same eigen values.
12. If λ is an eigen value of a non-singular matrix then λ^{-1} is an eigen value of A^{-1} .

Group-B

Answer any four questions

10×4

1. Find the angle between $\vec{A} = 2\hat{i} - 3\hat{j}$ and $\vec{B} = \hat{i} + 2\hat{j}$

Suppose a point P (5,2) is represented by **OP** in cartesian coordinates on the plan with $\hat{i}$ and $\hat{j}$ as standard base. Let $\hat{i}' = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$ and $\hat{j}' = \frac{-\hat{i}+\hat{j}}{\sqrt{2}}$ a new base. Write **OP** in terms of $\hat{i}'$ and $\hat{j}'$.

Show that $\vec{A} = (2\hat{i} - 2\hat{j} + \hat{k})/3$, $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})/3$ and $\vec{C} = (2\hat{i} + \hat{j} - 2\hat{k})/3$ are mutually orthogonal unit vector.

Find the shortest distance from (6, -4, 4) to the line joining (2, 1, 2) and (3, -1, 4).

2+3+3+2

2. If $\phi(x, y, z) = x^2yz$ and $\vec{A} = xz\hat{i} - xy^2\hat{j} + yz^2\hat{k}$, find $\frac{\partial^2(\phi\vec{A})}{\partial x\partial z}$ at the point (-2, 1, 1).

Evaluate $\nabla \cdot (\vec{r} \times \vec{A})$ if $(\vec{\nabla} \times \vec{A}) = 0$.

Discuss the physical significance of *divergence* and *curl* of a vector field with schematics.

3+3+4

3. Find the general solution of the differential equations

$$(D^2 + 1)y = x^2 \cos 2x$$

Construct the differential equation for damped vibration and find its displacement at any arbitrary time.

5+5

4. State and prove Cayley-Hamilton theorem.

If $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, Find a matrix P which diagonalizes A.

Verify that $P^{-1}AP = D$, and find A^6 .

Prove that roots of a Hermitian matrix are real.

2+5+3

5. Show that $\frac{d^n}{dx^n}(\delta(x)) = (-1)^n n! \frac{\delta(x)}{x^n}$

Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$

Deduce the expression for divergence in spherical co-ordinate system

3+3+4